Information for students

This coursework is intended to be a sample exam paper.

It accounts for 15% of the mark for this course.

Deadline: Friday, 5PM, 16 December, 2016. Pease submit a hard copy of your answers. A hand-written one is fine (typed answers also acceptable).

The Questions

1. Random variables.

a) The random variable X has a Gaussian distribution with mean 5 and standard deviation 2, and $Y = 2X + 4$. Find the mean, standard deviation and probability density function of Y.

[3]

b) The random variable X is uniform in the interval (0, 1). Find the density of the random variable $Y = -\ln X$.

[4]

c) X and Y are independent, identically distributed (i.i.d.) random variables with common probability density function

$$f_X(x) = e^{-x}, \qquad x > 0$$
$$f_Y(y) = e^{-y}, \qquad y > 0$$

Find the probability density function of the following random variables:

i) $Z = X + Y$. [6]

ii) $Z = \min(X, Y)$. [6]

iii) $Z = \max(X, Y)$. [6]

2. Estimation and sequences of random variables.

a) The random variable X has the density $f(x) \sim c^4 x^3 e^{-cx}$, $x > 0$. We observe the i.i.d. samples $x_i = 3.1, 3.4, 3.3$. Find the maximum-likelihood estimate of parameter c.

[8]

b) The random variable X_i are i.i.d. and uniform in the interval (0, 1). Show that if $Y = \max X_i$ then the distribution function $F(y) = y^n$ for $0 \leq y \leq 1$.

[8]

c) If the autocorrelation function $R_S(\tau) = Ie^{-|\tau|/T}$ and the linear MMSE estimate of $S(t - T/2)$ is given by $aS(t) + bS(t - T)$. Find the coefficients a and b and the corresponding mean-square error.

[9]

3. Random processes.

a) In the fair-coin experiment, we define the random process $X(t)$ as follows:

$X(t) = \sin \pi t$ if head shows;
$X(t) = 2t$ if tail shows.

i) Find the mean $E[X(t)]$. [2]

ii) Find the probability distribution function $F(x, t)$ for $t = 0.25$, $t = 0.5$, and $t = 1$. [3]

b) The random process $X(t)$ is real with autocorrelation $R(\tau)$.

i) Show that
$$P\{|X(t+\tau) - X(t)| \geq a\} \leq 2[R(0) - R(\tau)]/a^2.$$ [3]

ii) Express $P\{|X(t+\tau) - X(t)| \geq a\}$ in terms of the second order density $f(x_1, x_2; \tau)$ of $X(t)$. [2]

c) Suppose the wide-sense stationary random process $X(t)$ with $R_x(\tau) = 5\delta(\tau)$ passes through a linear system
$$Y'(t) + 2Y(t) = X(t)$$
Find $E[Y^2(t)]$, $R_{xy}(t_1, t_2)$, $R_{yy}(t_1, t_2)$ if the above equation holds for all t. [7]

d) Stochastic resonance. Suppose the input to the system with transfer function
$$H(s) = \frac{1}{s^2+2s+5}$$
is a wide-sense stationary process $X(t)$ with $E[X^2(t)] = 10$. Find the input power spectral density $S_x(\omega)$ such that the average power $E[Y^2(t)]$ of the resulting output $Y(t)$ is maximum.

[8]

4. Poisson processes and Markov chains.

a) Given a Poisson process $X(t)$, form its centered process $\tilde{X}(t) = X(t) - \lambda t$. Show that

$$E[\tilde{X}(t_1)\tilde{X}(t_2)\tilde{X}(t_3)] = \lambda \min(t_1, t_2, t_3)$$

[5]

b) Classify the states of the Markov chain with the following transition matrix

$$P = \begin{pmatrix} 0 & 1/2 & 1/2 \\ 1/2 & 0 & 1/2 \\ 1/2 & 1/2 & 0 \end{pmatrix}$$

[3]

$$P = \begin{pmatrix} 1/2 & 1/2 & 0 & 0 & 0 \\ 1/2 & 1/2 & 0 & 0 & 0 \\ 0 & 0 & 1/3 & 2/3 & 0 \\ 0 & 0 & 2/3 & 1/3 & 0 \\ 1/3 & 1/3 & 0 & 0 & 1/3 \end{pmatrix}$$

[5]

c) Consider the random walk in Lecture 9 with state space $E = \{0,1,2,\ldots\}$ and transition matrix

$$P = \begin{pmatrix} 0 & 1 & & & 0 \\ q & 0 & p & & \\ & q & 0 & p & \\ & & \ddots & \ddots & \ddots \\ 0 & & \ddots & \ddots & \ddots \end{pmatrix}$$

where $0 < p < 1$, $q = 1 - p$. Write a computer program to simulate the random walk and show the realizations of $X(t)$ as a function of t, for

i) $p = 1/3$; [3]

ii) $p = 1/2$; [3]

iii) $p = 2/3$. [3]

Discuss your findings. [3]

[Only show the simulation results as in Lecture 9 (not the program). Obviously, such a question cannot be tested in this way in the exam!]